

HEALTHCONNECT CLINIC EXPERIENCE LAB

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

WEEK 4 FOUNDATION AND DATA PREPARATION

In Week 4, the HealthConnect Data Analytics track focused on understanding appointment attendance and identifying factors associated with missed appointments. The main question was how HealthConnect could use data and AI to reduce missed appointments and improve patient support.

Week 5 moved from planning to practical analysis using the HealthConnect Appointment Dataset and Microsoft Power BI. The analysis considered patient demographics, appointment characteristics, booking behaviour, previous no-shows, reminders, distance to the clinic, waiting time, and appointment outcomes.

The dataset contained 5,000 appointment records and 18 variables, covering 1 January 2025 to 30 June 2026. Appointment IDs were unique, with no duplicate records identified.

Some missing values were found, including 90 distance-to-clinic values, 60 waiting-time values, and 1,366 reminder-channel values. The missing reminder-channel values were generally consistent with appointments where no reminder was sent.

A business-rule issue was also identified: the dataset contains Sunday appointments even though the Clinic Knowledge Base states that the clinic is closed on Sundays. This should be clarified before Sunday-related decisions are made.

EXPLORATORY ANALYSIS

The Week 5 analysis examined factors that may be associated with appointment no-shows, including:

- Previous no-shows
- Booking lead time
- Reminder status
- Appointment type
- Distance to the clinic
- Overall appointment outcomes

The dataset contained 5,000 appointments, consisting of:

- 2,423 No-Shows — 48.46%
- 2,314 Attended — 46.28%
- 263 Cancelled — 5.26%

The results show that no-shows are a major operational challenge because the no-show rate was higher than the attendance rate.

KPI DEVELOPMENT

KPI 1 — Total Appointments

Definition: The total number of appointment records in the HealthConnect Appointment Dataset.

Why It Matters: Total appointments provide a baseline for measuring the overall scale of appointment activity and interpreting the other appointment-performance KPIs.

Interpretation: HealthConnect recorded 5,000 appointments in the dataset, providing the overall population used for the analysis.

KPI 2 — Attendance Rate

Definition: The percentage of appointments that resulted in an attended appointment.

Business Question: What proportion of scheduled appointments were successfully attended?

Why It Matters: Attendance rate indicates how effectively scheduled appointments are being converted into completed patient visits.

Interpretation: The attendance rate of 46.28% indicates that fewer than half of the appointments in the dataset were attended.

KPI 3 — No-Show Rate

Definition: The percentage of appointments where the patient did not attend and did not cancel.

Business Question: What proportion of scheduled appointments were missed?

Why It Matters: No-show rate is one of the most important indicators for understanding unused appointment capacity and potential patient-engagement challenges.

Interpretation: The no-show rate of 48.46% is higher than the attendance rate of 46.28%, making missed appointments a major operational concern for HealthConnect.

KPI 4 — Cancellation Rate

Definition: The percentage of appointments that were cancelled.

Business Question: What proportion of scheduled appointments were cancelled?

Why It Matters: Cancellation rate helps HealthConnect understand how much appointment capacity is lost through cancellations and whether cancellation behaviour requires further investigation.

Interpretation: The cancellation rate of 5.26% is considerably lower than the no-show rate, indicating that missed appointments are a much larger issue than recorded cancellations in this dataset.

KPI 5 — Reminder Coverage

Definition: The percentage of appointments for which a reminder was sent.

Business Question: What proportion of appointments received a reminder?

Why It Matters: Reminder coverage measures the extent to which HealthConnect is using reminders as an appointment-support mechanism.

Interpretation: Reminder coverage of 72.68% means that reminders were sent for most appointments, but approximately 27.32% did not receive a reminder. This creates an opportunity to investigate whether additional appropriate reminder coverage could support attendance.

MAJOR BUSINESS INSIGHTS

Previous No-Shows:

Previous attendance behaviour showed the strongest pattern. The no-show rate increased from 43.51% for patients with no previous no-shows to 68.82% for patients with three or more previous no-shows.

Booking Lead Time:

Longer booking periods were associated with higher no-show rates. Appointments booked 31+ days in advance had a 60.49% no-show rate, compared with 26.51% for appointments booked 0–1 days in advance.

Reminder Status:

Appointments without reminders had a 51.39% no-show rate, compared with 47.36% when a reminder was sent. This suggests a potential relationship between reminder coverage and attendance, although it does not prove that reminders directly caused attendance.

Appointment Type:

Follow-up appointments had the highest no-show rate at 51.23%, followed by Diagnostic Tests at 49.75%, Specialist Consultations at 47.44%, and General Consultations at 46.64%.

Distance to Clinic:

Patients living more than 20 km from the clinic had a 57.76% no-show rate, suggesting that accessibility may be an important factor.

RECOMMENDATIONS

Based on the findings, HealthConnect should:

- Prioritise additional support for patients with previous no-shows, particularly those with three or more.
- Strengthen communication and reminders for appointments booked 31+ days in advance.
- Increase appropriate reminder coverage, which currently stands at 72.68%.
- Continue analysing reminder effectiveness before assuming a causal effect.
- Investigate possible accessibility challenges for patients living far from the clinic.
- Investigate why follow-up appointments have relatively high no-show rates.
- Resolve the Sunday operating-day inconsistency between the dataset and Clinic Knowledge Base.

6. Power BI Dashboard

The Week 5 Power BI dashboard provides a visual overview of appointment performance.

It contains five KPI cards:

- Total Appointments — 5K
- Attendance Rate — 46.28%
- No-Show Rate — 48.46%
- Cancellation Rate — 5.26%
- Reminder Coverage — 72.68%

It also includes visuals showing:

- No-Show Rate by Previous No-Shows
- No-Show Rate by Booking Lead Time
- No-Show Rate by Reminder Status

- No-Show Rate by Appointment Type
- No-Show Rate by Distance to Clinic
- Overall Appointment Outcomes

LIMITATIONS AND RISKS

The analysis has several limitations. Missing values exist in distance, waiting time, and reminder-channel fields. The Sunday operating-day issue also requires clarification.

Most importantly, the findings show associations, not causation. For example, the lower no-show rate among patients who received reminders does not prove that reminders caused patients to attend.

The dataset is also fictional and anonymised, so the findings should not automatically be generalised to real healthcare organisations.

COLLABORATION WITH DATA SCIENCE

The analysis identified several variables that could be considered for future predictive modelling, including:

- Previous no-shows
- Booking lead time
- Reminder status
- Reminder channel
- Appointment type
- Distance to clinic
- Other appointment characteristics

These variables can provide a foundation for future prediction of patients who may be at higher risk of missing appointments. However, they should first be assessed for data quality, predictive value, and possible data leakage.

WEEK 6 FOCUS

Week 6 should build on the Week 5 findings by:

- Conducting deeper analysis.
- Validating the strongest patterns.
- Refining the Power BI dashboard.
- Investigating high-risk appointment groups.
- Examining reminder effectiveness in greater detail.
- Resolving outstanding data-quality and business-rule issues.
- Supporting future predictive analysis.
- Providing further analytical inputs to the Data Science track.

CONCLUSION

Week 5 successfully moved the HealthConnect Data Analytics project from planning to practical implementation. Analysis of the 5,000-record HealthConnect Appointment Dataset produced five KPIs, a Power BI dashboard, and several important business insights.

The most significant issue is the 48.46% no-show rate, which is higher than the 46.28% attendance rate. Previous no-show history and booking lead time were the strongest patterns identified. Patients with 3+ previous no-shows had a 68.82% no-show rate, while appointments booked 31+ days in advance had a 60.49% no-show rate.

Reminder coverage, appointment type, and distance to the clinic also provide important areas for further investigation. However, these findings should be treated as associations rather than proof of causation.